Addition and subtraction of radicals

1

a $26\sqrt{3}$ b $-18\sqrt{5}$ c $27\sqrt{6}+11\sqrt{13}$ d $20\sqrt{6}+5\sqrt{11}$

e $4\sqrt{6}$ f $16\sqrt{5}$ g $-3\sqrt{2}$ h $42\sqrt{5}$

i $11\sqrt{10}$ j $-15\sqrt{5}$ k $28\sqrt{2}$ l $41\sqrt{3}+12\sqrt{7}$

m $10\sqrt{3}$ n $-8\sqrt{3}$ o $4+3\sqrt{5}$

2

a $3\sqrt{3}$ b $3\sqrt{3}$ c $5\sqrt{2}$ d $6\sqrt{5}$

3

a $3\sqrt[3]{v}$ b $7\sqrt[3]{u}$ c $2x^2\sqrt{ax}$

Multiplication and division of radicals

4

a $\sqrt{22}$ b $4\sqrt{2}$ c $20\sqrt{7}$ d $15\sqrt{35}$

e $\sqrt{70}$ f $35\sqrt{21}$ g $\sqrt{35}$ h $80\sqrt{5}$

i $\sqrt{231}$ j $11\sqrt{5}$ k $20\sqrt{11}$ l $30\sqrt{55}$

m $14\sqrt{11}$ n $24\sqrt{15}$ o $320\sqrt{3}$ p $40\sqrt{51}$

q $340\sqrt{7}$ r $216\sqrt{17}$

5

a $\sqrt{5}$ b $2\sqrt{7}$ c 32 d $8\sqrt{11}$

e $4\sqrt{7}$ f $\sqrt{3}$ g $\sqrt{11}$ h $\sqrt{3}$

i $\sqrt{7}$ j $\sqrt{13}$ k $5\sqrt{7}$ l $10\sqrt{5}$

m $15\sqrt{2}$ n $2\sqrt{7}$ o 3 p 6

q 10 r 16 s 10 t $\frac{5}{9}$

u $\frac{9}{2}$

6

a 2 b $\frac{1}{\sqrt{5}}$ c 4 d $\frac{1}{\sqrt{3}}$

e $\frac{1}{\sqrt{3}}$ f 2 g $\frac{2}{3}$ h $\frac{3}{2}$

i $17\sqrt{7}$

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7

a $4j\sqrt{k}$

b y^3

8

a $\frac{1}{7}$

b $\sqrt{\frac{p}{q}}$

c $3\sqrt{7}$

Mixed operations with radicals

9

a $-5\sqrt{7}+\sqrt{35}$

b $7\sqrt{14}+42\sqrt{2}$

c $-42\sqrt{22}+7\sqrt{77}$

d $\sqrt{77}+4\sqrt{11}$

e $3\sqrt{7}+\sqrt{21}$

f $-6\sqrt{2}+\sqrt{22}$

g $3\sqrt{39}-15\sqrt{3}$

h $\sqrt{33}+\sqrt{39}$

i $-4\sqrt{77}+4\sqrt{14}$

j $15\sqrt{11}+3\sqrt{55}$

k $-24\sqrt{14}+8\sqrt{6}$

l $15\sqrt{10}+20\sqrt{14}$

m $63\sqrt{5}$

n $11\sqrt{15}$

o $24\sqrt{77}-32\sqrt{55}$

p $\sqrt{143}+2\sqrt{13}$

q $-24\sqrt{55}-56\sqrt{77}$

10

a $\sqrt{26}-3\sqrt{13}+5\sqrt{2}-15$

b $\sqrt{14}-\sqrt{77}-\sqrt{26}+\sqrt{143}$

c $12+2\sqrt{11}$

d $29+12\sqrt{5}$

e $40-5\sqrt{3}-8\sqrt{7}+\sqrt{21}$

f $104-6\sqrt{55}$

g -110

h 3

i 25

11

a 60

b $\frac{12\sqrt{3}}{\sqrt{5}}$

c $\sqrt{5}+\sqrt{7}$

d $\frac{3\sqrt{3}}{\sqrt{10}}$

12

a $x=5999, y=-560$

b $x=5, y=-3$

13

a $42+(8+2m)\sqrt{13}+m^2$

b $m=-4$

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14

a u b $3u$ c $10\sqrt{u}$ d $6\sqrt{u}+\sqrt{v}$

e $15\sqrt{a}$ f $3\sqrt{5a}$

Rationalize the denominator

15

a No b Yes c No

16

a $\frac{\sqrt{11}}{11}$ b $\frac{\sqrt{22}-2\sqrt{11}}{11}$ c $\frac{7\sqrt{12}}{12}$ d $\frac{14\sqrt{143}}{117}$

e $-\frac{3\sqrt{13}}{65}$ f $7\sqrt{3}$ g $\frac{10\sqrt{35}-\sqrt{5}}{5}$ h $\frac{\sqrt{3}}{3}$

17

a $2-\sqrt{7}$ b $8+\sqrt{u}$ c $4+3\sqrt{10}$ d $-5\sqrt{3}-4\sqrt{v}$

18

a $\frac{-9\left(\sqrt{5}+7\right)}{44}$ b $4\left(\sqrt{10}+\sqrt{3}\right)$ c $\frac{5\sqrt{42}-4\sqrt{14}}{118}$ d $\frac{13+2\sqrt{22}}{9}$

e $\frac{887+144\sqrt{35}}{247}$ f $\frac{4\left(9\sqrt{11}+5\right)}{433}$ g $\frac{9+2\sqrt{14}}{5}$ h $\frac{\sqrt{10}-1}{9}$

19

a $\frac{6\sqrt{10}+5}{20}$ b $\frac{3\sqrt{5}-15\sqrt{2}}{20}$ c $-\frac{11\sqrt{15}}{18}$ d $-\frac{2\sqrt{2}}{23}$

20 No solution provided

21

a $\frac{\sqrt{3}-1}{2}$ b 16

22

a $19+8\sqrt{3}$ b $\frac{6-\sqrt{3}}{3}$

23 $\sqrt{1800}$

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Applications

24 $12\sqrt{2}$ units

25 $3\sqrt{5}$ units

26

a $4\sqrt{7}$ ft **b** 3 ft^2

27 $c = \sqrt{26}$ units

28 $9\sqrt{3}$ cm

29

a $\dfrac{\sqrt{5}+1}{2}$ **b** 1.618

30 $4\sqrt{10} - 12$ m

31

a 17 **b** $P = \dfrac{12\sqrt{x}+x}{5}$ **c** 17

32

a $\dfrac{\sqrt{3Vh}}{h}$ **b** 4 in

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